

Ming Huang

Software Architect | AI Systems & Data Engineering

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Software Architect and **AI Engineer** with 20+ years of experience building and re-engineering complex software systems, including 17 years in the UK. I'm also a **Content Creator** who makes AI engineering related (250+) videos for knowledge sharing.

Spent 11 years as a principal implementation authority for Panintelligence's multi-tenanted BI platform, architecting and implementing approximately 60% of the system across frontend and backend.

Now working in The Data City's research team on AI-assisted data processing at national scale, covering LLM evaluation, OCR and document-processing pipelines, industrial classification, summarisation, token and inference optimisation, MCP, and large-scale data engineering.

Strong focus on making AI systems practical in production: measuring model quality, throughput, reliability and cost rather than treating model capability in isolation.

SCALABLE LLM CODING ARCHITECTURE AND OPTIMISATION

Agent Harness Engineering: Designing agent execution so quality, reliability, speed and cost are controlled by the system, with multi-agent orchestration, runtime-enforced workflows and optimised model usage.

LLM-Maintainable Code Architecture: Structuring code so AI agents produce isolated, testable changes that remain consistently maintainable, with minimal refactoring and test rework as the system evolves.

Scalable Automation Architecture: Separating UI interaction from process flow so large Playwright automation systems remain maintainable as the target UI changes, with self-verifying, resilient and resumable workflows.

TECHNICAL EXPERTISE

Data & AI Pipelines: large-scale batch processing, OCR, document extraction, summarisation, embeddings, RAG, resumable and idempotent pipelines, DuckDB, Parquet, Qdrant, PostgreSQL, Neo4j

Agentic Systems: Claude Code, multi-agent workflows, MCP server development, runtime-enforced agent processes, concurrent worker architectures

Software Architecture: multi-tenanted SaaS, distributed and concurrent processing, API architecture, performance engineering, clean architecture, system refactoring

Languages: Java, Python, JavaScript, TypeScript, Groovy, C#, SQL, HTML, CSS

Frameworks & Infrastructure: Spring Boot, Grails, Angular, React, FastAPI, Pydantic, SQLAlchemy, Docker, Kubernetes, AWS, Cloudflare, Traefik

PROFESSIONAL EXPERIENCE

Developer – AI Integration

The Data City | Aug 2025 – Present

Work within the research team on AI-assisted processing of company and industrial data at national scale. Focus areas include LLM and OCR pipelines, classification, summarisation, model evaluation, document extraction, and the performance and cost engineering required to process millions of records.

- **PDF Text to Tables** - Built a scalable, traceable document-extraction pipeline (approximately 1.62 million pages) that progressively isolates sections and fields for deterministic extraction, with every decision auditable and LLMs reserved for lower-quality text - significantly improving accuracy, speed and cost at national scale.

- **Whole UK Investment Data Processing** - Converted an interactive Claude-based company-classification approach into a scalable Claude Code workflow, introducing concurrent workers, resumable processing and automatic recovery of abandoned work.
- **LLM Summary for Millions of Websites** - Engineered local-LLM summarisation for 1.2 million websites, reducing raw web text to ~33% through deterministic cleaning before inference and achieving ~1.5 seconds per website.
- **Model Evaluation & Prompt Engineering** - Evaluated models and engineered prompts for large-scale local inference, balancing output quality, schema reliability and throughput so a small non-reasoning model could handle the workload consistently at scale.
- **Resumable & Idempotent Pipelines** - Built resumable and idempotent AI pipelines where unchanged re-runs make zero model calls and return identical output.
- **MCP Integrations** - Created MCP solutions exposing internal APIs to AI-assisted workflows.
- **AI Infrastructure** - Built supporting AI infrastructure including FastAPI services, DuckDB-to-Qdrant embedding pipelines, Dockerised model services and internal HTTPS deployment through Cloudflare Tunnel and Traefik.
- **Similarity Search at Scale (Vector DB)** - Built a vector search system storing and querying 10M+ records, balancing storage size, build speed and query latency.
- **Smart Search at Scale (Micro Services)** - Built distributed search across 10M+ records with an Nginx load balancer, a fleet of FastAPI services, Redis caching and a database backend.

Independent AI Systems Development

Mar 2025 – Present

Independent research and development focused on practical AI engineering, agentic software development and lower-cost model operation.

- **Co-work Agent Team** - Built a multi-agent development framework supporting worker, reviewer, researcher and supervisory roles across a structured delivery lifecycle.
- **Runtime-Enforced Agent Workflows** - Enforced agent workflow and permissions through runtime hooks and tested code rather than relying solely on model instructions.
- **AI-Assisted Engineering Skills Library** - Developed a reusable library of AI-assisted engineering skills and methods covering architecture, data pipelines, UI automation, research and development workflows.
- **Local AI Development Environments** - Built and evaluated local AI development environments using Ollama and smaller language models, including automated capability testing across file operations, shell execution, parallel agents and data-processing tasks.
- **Media & AI-Generation Pipelines** - Developed reusable image, video and AI-generation pipelines with structured prompting and automated validation.
- **UK Tax Return Automation** - Built an end-to-end UK tax return automation pipeline covering document collection, normalisation, grouping, extraction, verification and report generation, with idempotent re-runs and verified accuracy.

AI Related Expertise

With a strategic focus on AI integration, I maintain comprehensive awareness of the rapidly evolving AI technology ecosystem. This knowledge enables me to make informed architectural decisions, identify optimal integration pathways, and advise on technology selection.

Category	Tools/Technologies
AI Tooling	Claude Code, Codex, Deepseek Harness
API Integration	OpenAI API, Anthropic API, Groq API, Ollama, vLLM
Vector Databases	Qdrant, Duckdb, Pinecone, Postgres
Workflows & Automation Integration	Coze, Flowise, n8n
AI Development Frameworks and RAG	LangChain
Model Context Protocol (MCP)	MCP Framework, Smithery (MCP Server Library)
Monitoring & Debugging	Grafana, Prometheus, Sentry
Cloud Platforms	Railway, Cloudflare, Supabase, AWS
AI Models	Hugging Face

Knowledge Sharing / Content Creator (+250 technical video)

Created 250+ technical videos focused on AI integration, software engineering and practical AI implementation, building an audience of 6,000+ subscribers. Refer to my personal site for further info.

Technical Architect

[Panintelligence](#) | Mar 2025 – Jul 2025

- Built and published an MCP server for the Panintelligence API in March 2025.
- Developed AI-assisted data insight capabilities using the platform's existing multi-tenanted data-security model.
- Designed the architecture for AI-generated data summaries and insights.
- Integrated AI-driven theme generation into the platform's existing theming architecture.

Principal Software Engineer

[Panintelligence](#) | Nov 2021 – Mar 2025

Principal implementation authority for the platform, responsible for complex work requiring detailed knowledge across frontend, backend, database and product architecture.

- Architected and delivered the transformation of the platform from a single-organisation system to a **multi-tenanted architecture**, spanning database, API, UI and data migration.
- Designed and implemented the platform's v2 REST APIs, with documentation generated directly from code metadata to maintain alignment between implementation and API documentation.
- Led backend scalability, caching and SQL-performance optimisation for high-concurrency environments.
- Built complex reusable platform components, including a cross-browser text-editing component that remained stable in production for more than three years.
- Reviewed architecture and implementation across the engineering team and mentored developers on solution design.
- Established architectural foundations for AI integration using RAG and vector databases.

Software Architect

[Panintelligence](#) | Mar 2015 – Nov 2021

Led major frontend and platform re-engineering for the Panintelligence Responsive HTML5 dashboard across desktop, tablet and mobile.

- Designed a high-performance reactive Angular architecture using functional and immutable data patterns.
- Built an IDE-style configuration interface supporting simultaneous editing of users, roles and data connections with independent undo/redo.
- Established frontend architecture, reusable component patterns and design guidelines used by the wider development team.

- Designed URL-driven application state and navigation.
- Built the frontend entity-relationship configuration interface and backend SQL-generation engine, including dynamic query generation and multi-level data security.
- Designed the platform's theming architecture, later reused for AI-generated themes.

Public Speaker

- Presented technical work at the Panintelligence User Conference, Kirona User Event and Agile Yorkshire.
- Agile Yorkshire Speaker - Making Use of Data Visualisation for Software Development

Software Engineer

Panintelligence | Jul 2014 – Mar 2015

- Introduced Material Design principles into product development.
- Established the Gradle build and dependency-management process.
- Improved frontend architecture using object-oriented JavaScript, BEM and SMACSS.
- Evaluated and introduced technologies including Grails and D3.

Analyst Programmer

CPP Group plc | Dec 2007 – Jul 2014

Full-stack development for financial systems supporting credit-card insurance products.

- Re-engineered a live web application from Foundation to Bootstrap, reducing approximately **6,000+ lines of CSS by half** while maintaining cross-browser compatibility.
- Developed online purchase and customer-servicing systems using J2EE, Spring, Hibernate, Groovy and Grails.
- Led code-review practices and frontend quality standards within the development team.

EDUCATION

MSc Software Engineering

University of York